

Lecture Notes on Differential Equations and Transform Techniques

Lecture Notes for Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.

EE1060: Differential Equations and Transform Techniques**Lecture Notes - 04****Topics covered :**

1. First-order ODE - Seperable Equations
2. First-order ODE - Exact Equation

- A first-order differential equation [1]

$$\frac{dy}{dx} = -\frac{P(x, y)}{Q(x, y)} \quad (1)$$

is called a **separable** equation if $\frac{P(x, y)}{Q(x, y)}$ can be written as a product of a function of x and a function of y i.e.,

$$\frac{dy}{dx} = f(x)g(y) = \frac{f(x)}{g(y)} \quad (2)$$

It is also a common practice to write the DE of the form (1) as

$$P(x, y)dy + Q(x, y)dx = 0 \quad (3)$$

The solution $y = \phi(x)$ to a separable differential equation of the form (2) can be obtained by integrating both sides with respect to the independent variable x (or more generally, the independent variable), as follows.

$$\int g(y) \frac{dy}{dx} dx = \int f(x) dx + c \quad (4)$$

This leads to the following result ¹

$$\int g(y) dy = \int f(x) dx + c \quad (5)$$

- Let $f(x, y) = 0$ be an implicit function. Differentiating $f(x, y)$ implicitly with respect to x gives

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \frac{dy}{dx} = 0 \quad (6)$$

Thus, an implicit function $f(x, y) = 0$ is a solution to a differential equation $P(x, y)dx + Q(x, y)dy = 0$ if $\frac{\partial f}{\partial x} = P(x, y)$ and $Q(x, y) = \frac{\partial f}{\partial y}$. This relationship forms the basis of an exact differential equation.

- A first-order ODE $P(x, y)dx + Q(x, y)dy = 0$ is **exact** in a region R if there exists a function $f(x, y) = 0$ such that

$$\frac{\partial f}{\partial x} = P(x, y) \text{ and } \frac{\partial f}{\partial y} = Q(x, y) \quad (7)$$

for all $(x, y) \in R$.

¹Recall the idea of differentials

- **Necessary and Sufficient Condition for an Exact Differential Equation [1]:** A first-order differential equation $P(x, y) dx + Q(x, y) dy = 0$ where $P(x, y)$ and $Q(x, y)$ are continuously differentiable functions in a Region R is exact if and only if

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \quad (8)$$

at every point in R .

Proof of Necessity: Suppose the differential equation is exact. This means that there exists a function $f(x, y)$ such that $\frac{\partial f}{\partial x} = P(x, y)$ and $\frac{\partial f}{\partial y} = Q(x, y)$.

Differentiating $P(x, y) = \frac{\partial f}{\partial x}$ with respect to y and $Q(x, y) = \frac{\partial f}{\partial y}$ with respect to x results in

$$\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial x \partial y} \quad (9)$$

and

$$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial x \partial y} \quad (10)$$

Since the partial derivatives are continuous (implying that the mixed second-order partial derivatives are continuous), we have (by the Schwarz theorem)

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} \quad (11)$$

implying that

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \quad (12)$$

Equation (12) establishes the necessary part of the proof.

Proof of Sufficiency: To prove the sufficiency, we need to show that if $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$, then there exists a function $f(x, y)$ such that $\frac{\partial f}{\partial x} = P(x, y)$ and $\frac{\partial f}{\partial y} = Q(x, y)$. To check if such a function exists (or more precisely, can be constructed), we first observe that the function (13) satisfies the condition $\frac{\partial f(x, y)}{\partial x} = P(x, y)$.

$$f(x, y) = \int P(x, y) dx + g(y) \quad (13)$$

where $g(y)$ is an arbitrary function of y (and y alone).

Now, to proceed, we need to show that we can choose the arbitrary function $g(y)$ such that $\frac{\partial f}{\partial y} = Q(x, y)$ and that this function depends only on y . To achieve this, we differentiate the expression (13) with respect to y and set it equal to $Q(x, y)$. This results in

$$g'(y) = Q(x, y) - \frac{\partial}{\partial y} \int P(x, y) dx \quad (14)$$

Note that for $g'(y)$ (given by (14)) to be a function of y alone, it is sufficient to show that its partial derivative

with respect to x is zero. This condition is met if

$$\frac{\partial g'(y)}{\partial x} = \frac{\partial Q(x, y)}{\partial x} - \frac{\partial}{\partial x} \frac{\partial}{\partial y} \int P(x, y) dx = \frac{\partial Q(x, y)}{\partial x} - \frac{\partial}{\partial y} \frac{\partial}{\partial x} \int P(x, y) dx = \frac{\partial Q(x, y)}{\partial x} - \frac{\partial P(x, y)}{\partial y} = 0 \quad (15)$$

Since $g'(y)$ does not depend on x , it follows that $g(y)$ itself is also independent of x . In other words $g(y)$ is purely a function of t alone.

Equation (15) shows that if $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$, a function $f(x, y)$ that satisfies $\frac{\partial f}{\partial x} = P(x, y)$ and $\frac{\partial f}{\partial y} = Q(x, y)$ can be constructed, which implies that differential equation is exact. This completes the sufficiency part of the proof.

- **Example:** Check if the ODE (16) is exact. If it is, compute its general solution.

$$y^n dx + nxy^{n-1} dy = 0 \text{ where } n > 1 \quad (16)$$

For the given ODE, $P(x, y) = y^n$ and $Q(x, y) = nxy^{n-1}$. The partial derivatives of $P(x, y)$ with respect to y and partial derivative of $Q(x, y)$ with respect to x are

$$\frac{\partial P(x, y)}{\partial y} = ny^{n-1} \text{ and } \frac{\partial Q(x, y)}{\partial x} = ny^{n-1} \quad (17)$$

Since $\frac{\partial P(x, y)}{\partial y} = \frac{\partial Q(x, y)}{\partial x}$, the given differential equation is exact. Notably, (16) can also be rewritten as $y' = -\frac{y}{nx}$, or equivalently, as

$$y dx + nxy dy = 0 \quad (18)$$

While (18) and (16) appear to be equivalent, the former is not exact, whereas the latter is an exact differential equation.

To determine the general solution of (16), we first integrate $P(x, y)$ with respect to x , yielding

$$f(x, y) = \int P(x, y) dx + g(y) \implies f(x, y) = xy^n + g(y) \quad (19)$$

where $g(y)$ is an arbitrary function. Equation (19) represents a solution if $\frac{\partial f(x, y)}{\partial y} = Q(x, y)$. To satisfy this condition, $g(y)$ must fulfill

$$nxy^{n-1} + \frac{\partial g(y)}{\partial y} = nxy^{n-1} \implies g'(y) = 0 \implies g(y) = c \quad (20)$$

Thus, the general solution of ODE (16) is $f(x, y) = xy^n + c = 0$, where c is an arbitrary constant.

References

- [1] C. H. Edwards and D. E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th ed. Upper Saddle River, NJ: Pearson, 2000.